G2 · WIRING

G2 Wiring Guide

Every wire and contact inside G2. Part 1 covers the power and brain stack: PiSugar under the Pi, and the Pi wired to the BiBoard. Part 2 covers everything else wired into the Pi: the microphone and amplifier, and the Petoi AI head.

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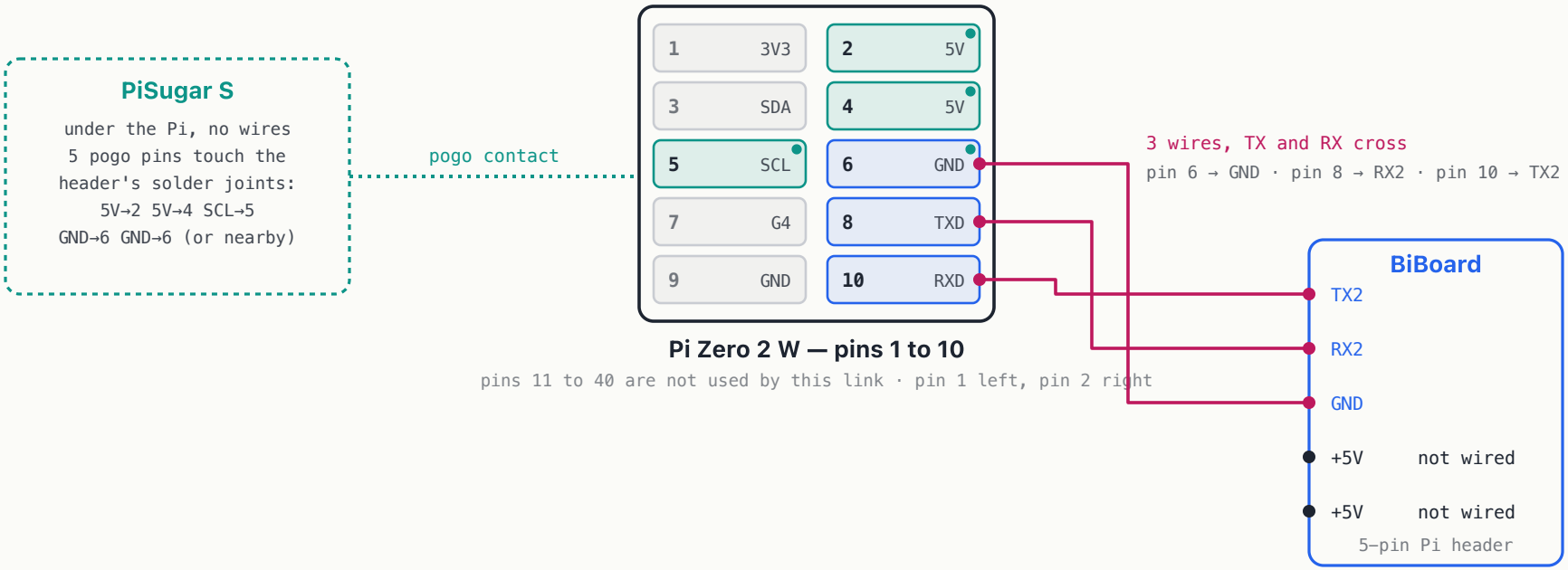
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PART 1

PiSugar, Pi and BiBoard

PiSugar powers the Pi through contact pins, and the Pi talks to the BiBoard over a three-wire serial link.

1.1 Stack and power



PiSugar powers the Pi through its pogo pins, so the BiBoard's +5V pins and the Pi's pin 2 carry no supply for this link.

1.2 PiSugar to Pi (contact, no wires)

- Four screws hold PiSugar under the Pi. The holes are symmetric, so it bolts on in two rotations and only one powers the Pi.
- **Correct:** PiSugar's micro-USB port, switch and battery end up under the Pi's mini-HDMI and USB edge, and its pogo pins land under the 40-pin header.
- Check that the pogo pins sit on the header's solder joints before tightening. The screws fit either way.

POGO PIN	PI PIN (INFERRED)
5V	2
5V	4
SCL (auto-start)	5
GND	6
GND	6 or nearby

The 2/4/5/6 range is confirmed by PiSugar's PCB legend and FAQ. The pin-by-pin match is not independently verified.

1.3 Pi to BiBoard (three wires, crossed)

BiBoard's 5-pin Pi header sits at the head-facing edge, above the G1 Grove socket. Left to right: TX2 (square pad), RX2, GND, +5V, +5V. Solder a 5-pin header there if none is fitted.

BIBOARD	PI PIN	SIGNAL
TX2	10	GPIO15 · RXD
RX2	8	GPIO14 · TXD
GND	6	GND
+5V, +5V	not wired, PiSugar powers the Pi	

TX and RX cross: each board's transmit line goes to the other's receive line. Pi TXD (pin 8) carries the Pi's outgoing data to BiBoard RX2, and BiBoard TX2 goes to Pi RXD (pin 10). Ground runs straight across. Use jumpers long enough to reach the header at the final mounting position (20 to 30 cm if the Pi sits at the back edge).

1.4 Before power-on

- Both of BiBoard's +5V pins are empty.
- No BiBoard wire is on the Pi's pin 2. (Pin 2 does carry the amp's 5 V in Part 2; that is a separate wire.)
- BiBoard TX2 goes to Pi **pin 10**, not pin 8. They are easy to swap.
- All three wires are seated at both ends.

PART 2

Everything wired into the Pi

Once Part 1 is built, the Pi's 40-pin header also feeds the audio pair. The AI head connects to the BiBoard by its Grove cable and to the Pi over Wi-Fi, so it adds no wires to the header.

2.1 Header map and wire map

Pin 1 top-left, pin 2 top-right, running down to 39 and 40. Odd pins left, even pins right. A teal ring marks pins PiSugar also touches.

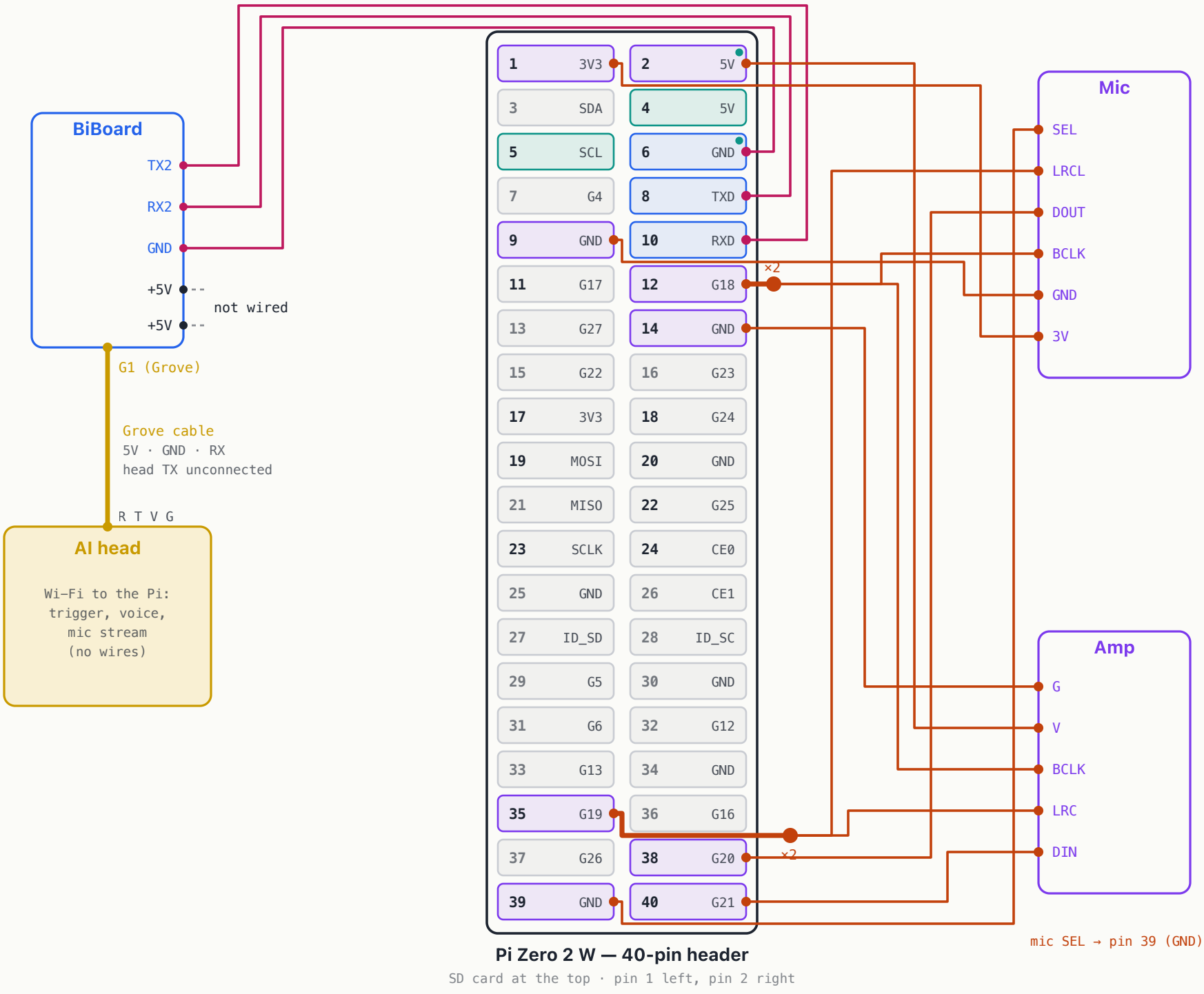
BiBoard Mic / amp PiSugar Free

1 3V3 Mic VIN	5V **2** Amp VIN · PiSugar 5V
3 GPIO2 · SDA *free*	5V **4** PiSugar contact
5 GPIO3 · SCL PiSugar auto-start tap	GND **6** BiBoard GND
7 GPIO4 *free*	GPIO14 · TXD **8** BiBoard RX2
9 GND Mic or Amp GND	GPIO15 · RXD **10** BiBoard TX2
11 GPIO17 *free*	GPIO18 **12** BCLK, splice to mic and amp
13 GPIO27 *free*	GND **14** *free*
15 GPIO22 *free*	GPIO23 **16** *free*
17 3V3 *free*	GPIO24 **18** *free*
19 GPIO10 · MOSI *free*	GND **20** *free*
21 GPIO9 · MISO *free*	GPIO25 **22** *free*
23 GPIO11 · SCLK *free*	GPIO8 · CE0 **24** *free*
25 GND *free*	GPIO7 · CE1 **26** *free*
27 ID_SD	ID_SC **28**

reserved, HAT EEPROM	reserved, HAT EEPROM
29 GPIO5 *free*	GND **30** *free*
31 GPIO6 *free*	GPIO12 **32** *free*
33 GPIO13 *free*	GND **34** *free*
35 GPIO19 LRCLK, splice to mic and amp	GPIO16 **36** Amp CTRL: no connection needed
37 GPIO26 *free*	GPIO20 **38** Mic DOUT
39 GND Mic SEL (tied to ground)	GPIO21 **40** Amp DIN

Wire map: how everything connects

The header drawn standing up, with each wire traced to the board it feeds. Mic and amp wires are orange and BiBoard wires are pink, so both stand apart from the purple and blue pin squares.



Pi Zero 2 W — 40-pin header

SD card at the top · pin 1 left, pin 2 right

Pin squares are colored by owner; pins 2, 4, 5 and 6 also touch PiSugar's pogo contacts (teal dot or fill). BiBoard's three wires land on pins 6, 8 and 10 in the right column, so they loop over the top of the header to reach it. The Pi drives BCLK and LRCLK only from pins 12 and 35, so each feeds a splice (thick wire, dot, ×2) that fans out to both the mic and the amp. Amp G is shown on pin 14, but any free GND pin works. The mic's SEL pin is tied to ground with a wire to pin 39 (a GND pin) — it is a fixed channel-select level, not a data signal. Board ports are in the order printed on each board (mic top to bottom as read; amp in plug order), so wires cross where those orders disagree with the header's pin order. The AI head (yellow) connects to BiBoard G1 by its four-pin Grove cable; its Wi-Fi links (trigger, voice, mic stream) need no wire.

2.2 Mic and amp (I2S)

BOARD PIN	PI PIN	SIGNAL
Mic VIN	1	3V3 only
Amp VIN	2	5V
Mic GND, Amp GND	9 or any free GND	GND
BCLK (both)	12	GPIO18, spliced
LRCLK (both)	35	GPIO19, spliced
Mic DOUT	38	GPIO20
Mic SEL	39	GND, left channel
Amp DIN	40	GPIO21
Amp CTRL	36	no connection needed

BCLK / LRCLK splice

The Pi drives these clocks only from GPIO18 and GPIO19, and both boards only listen, so each signal fans out to two boards. Build each splitter from three jumper wires joined at one soldered joint away from either board:

- female-to-female wire: female end to the Pi pin
- female-to-female wire: female end to the mic's header pin
- female-to-male wire: male end into the amp cable's matching wire

Strip the other ends, twist, solder, insulate with heat-shrink or tape. GND and VIN do not fan out; each board grounds and powers independently.

Mic board

- SPH0645 I2S breakout with a single row of 6 through-hole pads: 3V, GND, BCLK, DOUT, LRCL, SEL. It has no header, so solder a 6-pin 0.1 in (2.54 mm) male header into the pads.
- VIN runs at 1.6 to 3.6 V only. Connect it to a 3.3 V pin, never the 5 V pin.
- SEL must be tied to a fixed level. It runs to pin 39 (GND), so the mic speaks on the left channel only and the right channel records silence. Capture software reads the left channel.
- After soldering, check each pair of neighboring pins for continuity. A solder bridge makes every sample read the same value.

Amp board

- NS4168 I2S amplifier with a 5-pin PH2.0 connector silkscreened G, V, BCLK, LRCLK, DIN.
- The included cable is five separate wires, each ending in its own female socket. Leave it intact; the BCLK and LRCLK wires take the splice joints.
- The speaker plugs into the amp's 2-pin PH2.0 output. This is a separate circuit from the Pi wiring.
- Pin 36 (CTRL): the board shows no separate control pin, so nothing is wired.

Notes

"Fan-out" vs "either pin" — two different things look similar in the table but aren't. **GND** is flexible: every GND pin on the Pi is the same electrical net, so mic and amp can each use a different free ground pin — nothing forces them onto pin 9 specifically. **BCLK/LRCLK are not flexible**: the Pi's I2S hardware only drives those clocks out of GPIO18/GPIO19 specifically, and both boards only ever listen — they never drive the line themselves. One driver with two listeners isn't a conflict, but it does mean both boards physically need to reach that exact pin.

Pin 2 carries two things — PiSugar feeds 5V in here, and the amp draws 5V from the same pin. That's normal — it's a shared power rail, not a conflict — but when the amp's VIN wire is unplugged, a live pin 2 does not mean the amp is still connected.

2.3 Petoi AI head

LINK	PATH	WIRES
Power and serial	Head's 4-pin R T V G connector to BiBoard G1 (Grove, UART2)	4-wire Grove cable. The head's transmit line is left unconnected, so the Pi stays the only sender.
Trigger, voice, mic stream	Pi to head over Wi-Fi	None

G1 is the only UART-capable Grove socket on BiBoard V1 (G2 is I²C, G3 and G4 are analog).

2.4 Open items

- **One UART, two listeners.** The Pi's 5-pin header and the head's Grove socket both land on BiBoard UART2 (G1). Leaving the head's transmit line unconnected keeps the Pi as the only sender. Confirm which physical socket on the board is G1 before connecting the head.
- **Sharing option (Petoi, 2026-10-15):** the BiBoard's onboard voice command module can be disconnected and its port assigned to the head, so the head no longer shares UART2 with the Pi. The head can also take its 5 V from the BiBoard. Head UART pins are fixed in its firmware (`second_uart.h`).
- **Unmeasured:** the head's current on the Grove 5 V (BiBoard input is 7 to 9 V).